

- (a) Describe a method they could use to find the density of the metal. [6 QER]

(b) The table below gives data on the density of some common metals.

Metal	Density (g/cm ³)
aluminium	2.70
copper	8.96
gold	19.32
iron	7.87
tin	7.26

- (i) Rhys and Elliot calculate the density of their metal to be 8.1 g/cm³.
State which metal the irregular shape is most likely to be. Give a reason for your answer. [2]

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- (ii) Rhys and Elliot are not confident that they can correctly identify the metal. Suggest why they think this. [1]

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- (iii) The table below shows their results.

Mass (g)	Volume (cm ³)	Density (g/cm ³)
65	8	8.1

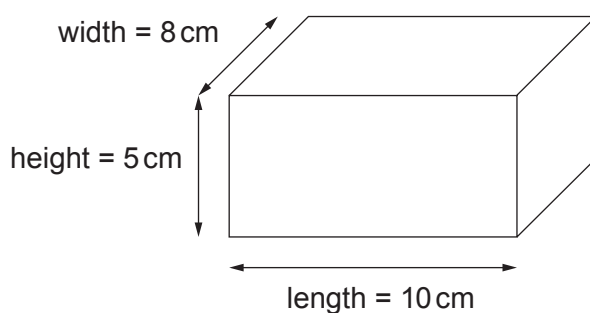
Suggest how Rhys and Elliot could get a more accurate value for the density. [1]

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- (c) The boys notice that gold has a high density of 19.32 g/cm^3 and they are interested in the mass of a gold block.



- (i) Use the equation:

$$\text{volume} = \text{length} \times \text{height} \times \text{width}$$

to calculate the volume of the gold block shown above.

[1]

Volume = cm^3

- (ii) Use the equation:

$$\text{mass} = \text{density} \times \text{volume}$$

to calculate the mass of the gold block.

[2]

Mass = g

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